expert knowledge) not found in public training sets. Integrating this specialized data enables complex tasks, such as:

- Analyzing turbine vibration readings.
- Cross-referencing them with the specific maintenance schedule and operating hours.
- Generating a detailed diagnostic report predicting a potential bearing failure.

The LLM thus moves beyond simple information retrieval to become an expert system capable of complex analysis and predictive recommendations.

5.3 Ontologies as a blueprint for understanding

To create accurate digital models, computers require a formal structure to organize and define concepts. An *ontology* is a formal, structured framework that defines the essential concepts, relationships, and properties within a given domain. Acting as a blueprint or schema, it precisely maps entities and their interconnections. This formalization allows machines to move beyond simple data processing to reason and infer, transforming raw data into a coherent knowledge map. This approach ensures the clarity, consistency, and shared vocabulary vital for building robust, scalable, and intelligent digital twin systems.

5.3.1 Core components of an ontology for a digital twin

The primary purpose of an ontology in a digital twin is to provide a comprehensive and structured knowledge model of a physical asset, system, or process. It acts as the semantic foundation that enables the digital twin to accurately represent, reason about, and interact with its real-world counterpart. At its core, a digital twin ontology is composed of the following key elements:

- *Concepts and classes* represent the types of entities that exist within the domain. In a digital twin of a factory for example, these might include `Machine`, `Product`, and `Building`.
- *Properties and attributes* define the characteristics of a concept providing the details that describe a specific instance. For example, a